

Customer Churn Prediction & Revenue Risk Analysis

Objective: Identify churn drivers and prevent revenue loss

5,000

DATASET

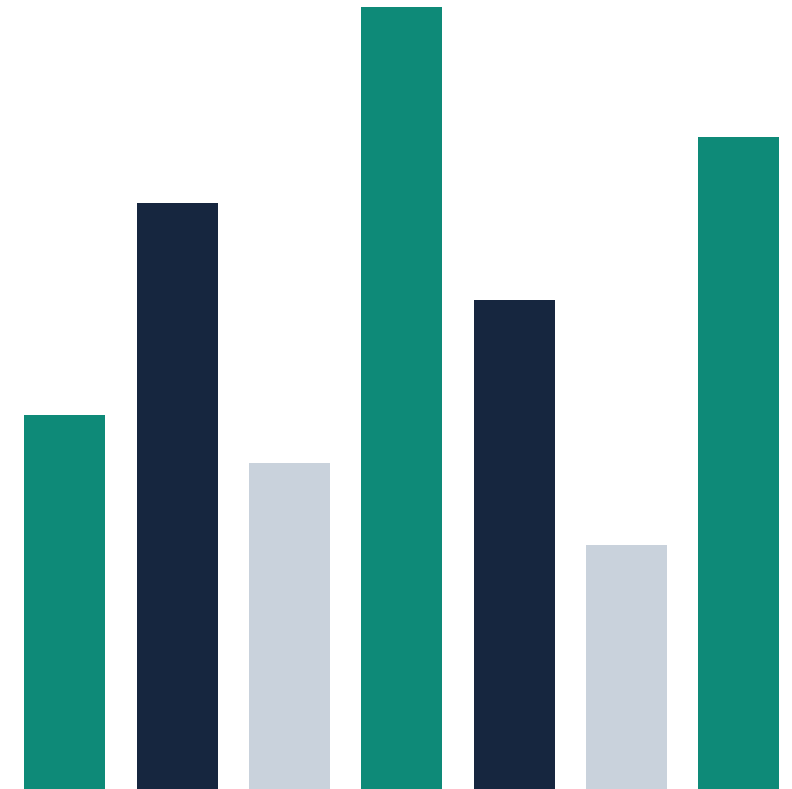
customers analyzed

Risk list

OUTPUT

who to call today

Models: Logistic Regression for churn risk; XGBoost for next-30-day prediction



WHY THIS MATTERS

Business Problem

Customer churn is not only a model outcome — it is recurring revenue leaving the business.

CHURN

Revenue loss

At-risk customers reduce monthly recurring revenue.

PREDICTION

Risk detection

Estimate who is likely to churn before renewal.

PRIORITIZATION

Retention action

Focus teams on the highest-value, highest-risk accounts.

Goal



Predict churn



Understand why

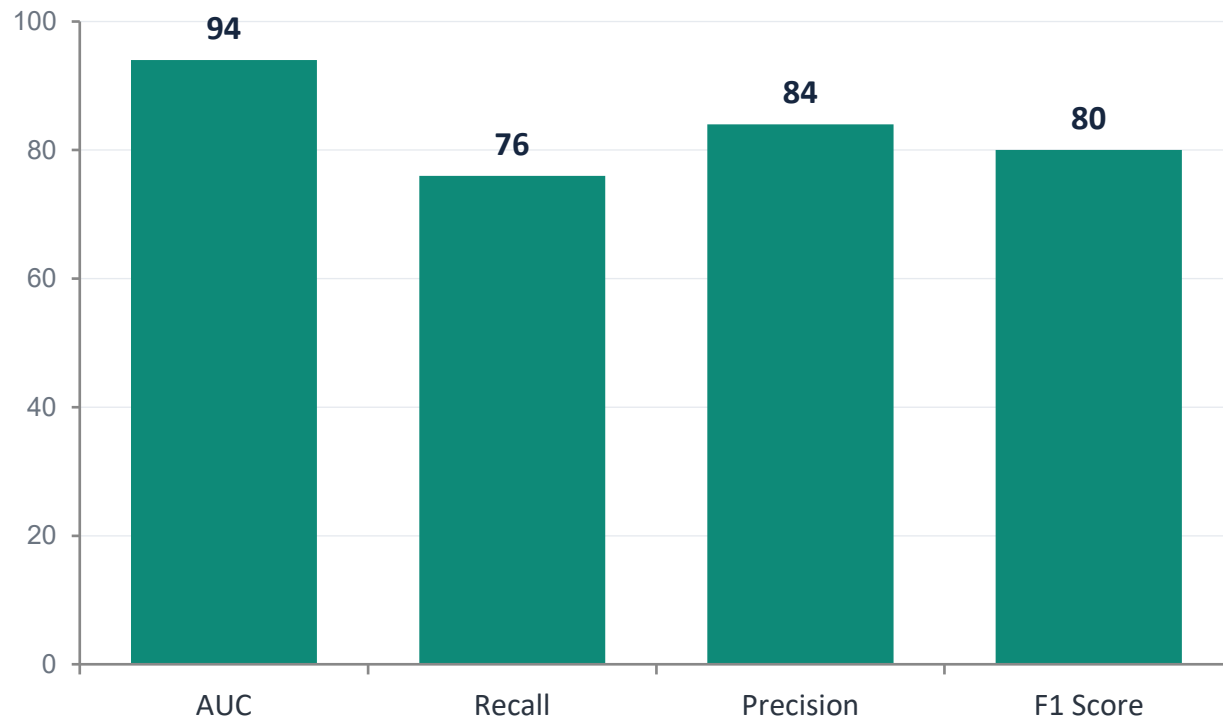


Prioritize actions

Management question: Which customers should we intervene on first to protect revenue?

Model Performance: Logistic Regression

The baseline churn model is strong enough to support operational prioritization.



0.94

AUC

excellent discrimination

84%

PRECISION

predictions are reliable

76%

RECALL

catches most churners

0.80

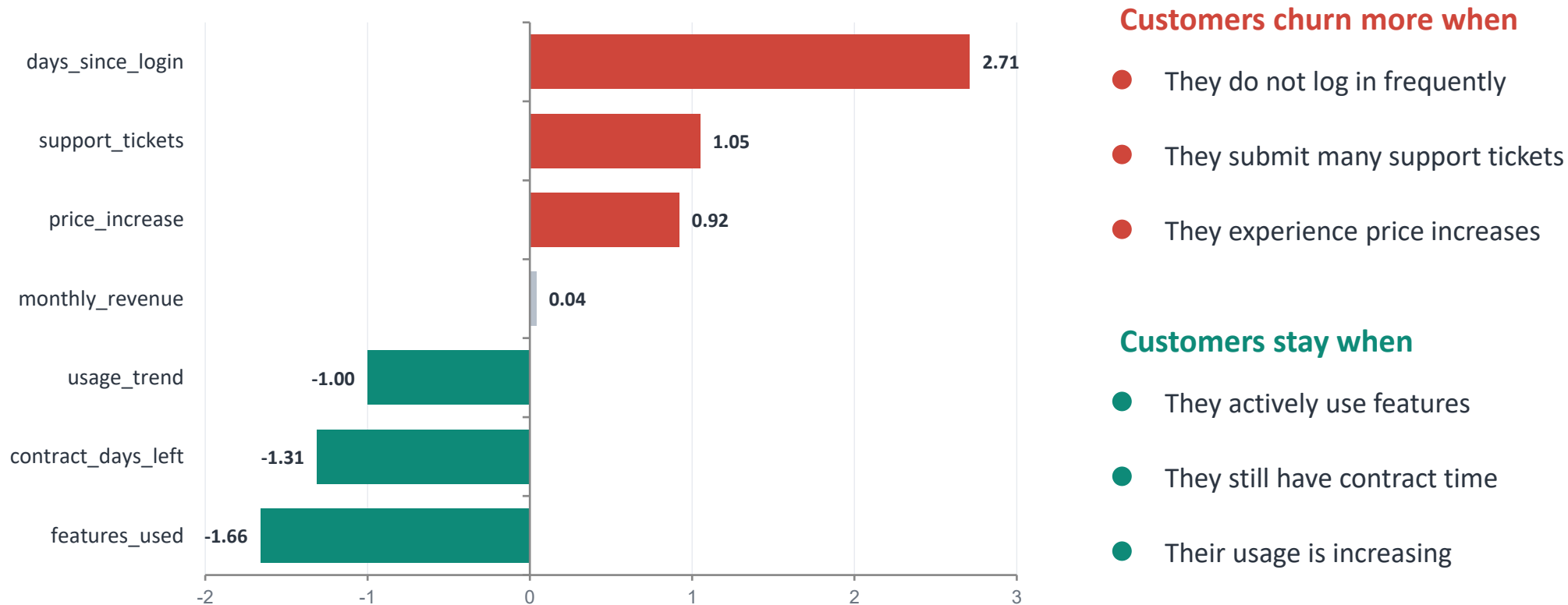
F1 SCORE

balanced performance

Interpretation: The model is highly accurate and reliable for identifying churn risk.

Why Customers Churn

The model highlights behavior and experience signals that management can act on.



This turns the model into business insight: engagement, support experience, and pricing are the levers.

Feature impact = logistic-regression coefficient (log-odds); positive increases churn risk, negative protects

REVENUE EXPOSURE

Financial Risk

At a 75% churn-probability threshold, the model isolates the highest-confidence revenue risk.

990

HIGH-RISK CUSTOMERS

>75% churn probability

\$422,296

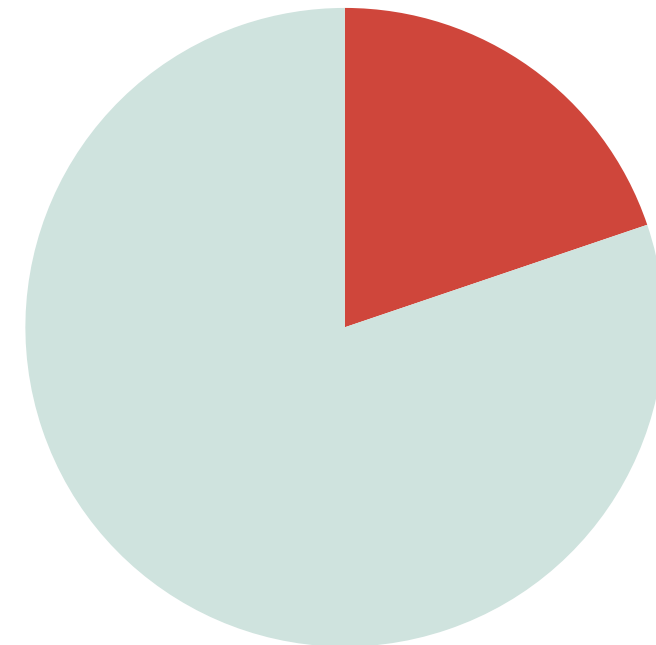
MONTHLY REVENUE AT RISK

prioritized retention pool

75% threshold

Focuses teams on the highest-confidence churners — fewer false alarms, faster action, clearer ownership.

High-risk segment = churn probability greater than 75%; safe / monitor = remaining customer base



■ High-risk ■ Safe / monitor

19.8% of customers are high risk

Who to Call TODAY

We should immediately contact high-value customers with the highest churn risk.

Customer ID	Churn Probability	Monthly Revenue	Priority Score
CUST_3924	99.99%	\$797	797
CUST_0551	99.51%	\$797	793
CUST_2922	99.74%	\$795	793
CUST_1844	99.82%	\$792	791
CUST_2701	99.29%	\$795	790
CUST_2109	99.69%	\$790	788
CUST_3428	98.64%	\$792	782
CUST_1103	98.93%	\$787	779
CUST_0843	98.63%	\$787	776
CUST_2503	98.57%	\$784	773

Priority score

Combines churn probability and monthly revenue (probability x revenue) to focus CS on the largest protection opportunities.

Action rule

Call today if churn probability exceeds 75% and revenue is top-tier.

Future Risk Customers

These customers are highly likely to churn in the next 30 days — proactive retention required.

Customer ID	Prob.	Revenue
C-02174	99.8%	\$789
C-05513	99.8%	\$451
C-03210	99.7%	\$787
C-04692	99.7%	\$446
C-01845	99.7%	\$373
C-06031	99.6%	\$335
C-00782	99.6%	\$328
C-04451	99.6%	\$316
C-02964	99.5%	\$289
C-01508	99.5%	\$179

Customer ID	Prob.	Revenue
C-05177	99.5%	\$451
C-02459	99.4%	\$410
C-03628	99.3%	\$372
C-04003	99.2%	\$355
C-00994	99.1%	\$338
C-05820	99.0%	\$305
C-01346	98.8%	\$284
C-03185	98.6%	\$262
C-04706	98.3%	\$233
C-02671	98.0%	\$208

Retention posture

Prioritize immediate outreach for the highest-probability customers and track daily response signals.

Top 20 shown

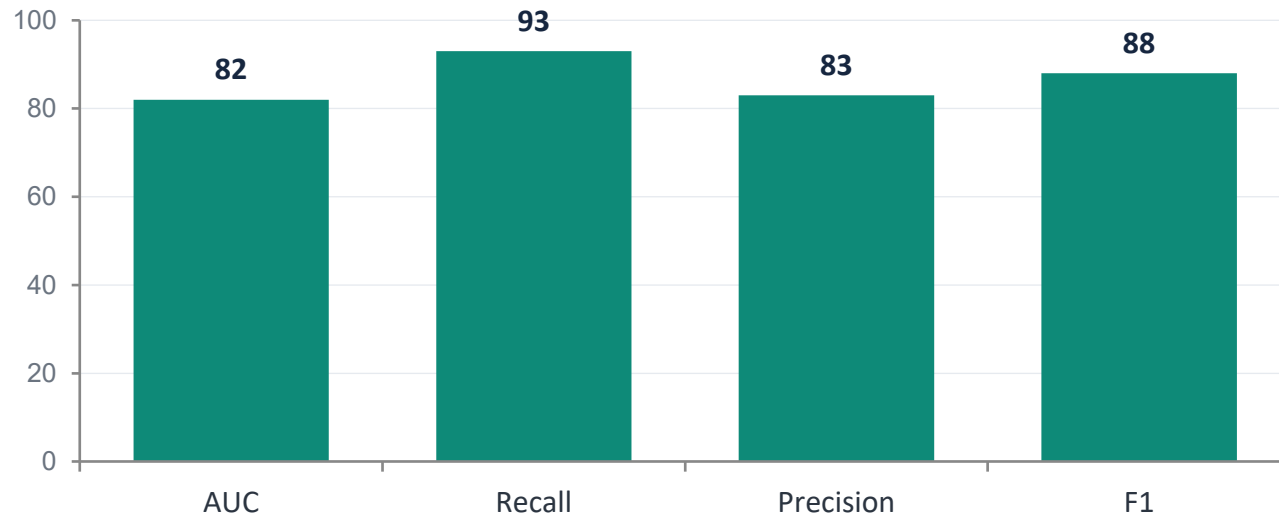
Sorted by predicted churn probability for the coming 30 days.

Placeholder rows at production scale — replace IDs and values with the live 30-day XGBoost export before outreach.

FUTURE RISK MODEL

30-Day Churn Prediction: XGBoost

For the next 30 days, the priority is catching almost all future churners — high recall by design.



Key message

“We prioritize catching almost all future churners.”

Operationally, the model is tuned for proactive retention, even if some lower-risk customers are included in outreach.

Priority Strategy

Use churn probability and revenue value to route customers into the right retention motion.

Risk segment	Action	Operational owner
High churn + high revenue	Immediate call	CS manager within 24 hours
High churn + low revenue	Email automation	Lifecycle campaign this week
Low churn	Monitor	Track engagement and renewals

VIP CHURN RISK = high risk + high revenue

Recommendation

Reducing support-ticket volume could significantly reduce churn risk, especially among high-value customers already showing low engagement.

Key Business Insights

Churn is predictable, explainable, and financially material — the next step is to take targeted retention action.

Top churn drivers

- **Low engagement** (days_since_login)
- **Poor experience** (support tickets)
- **Pricing changes**

Revenue risk

\$422K

monthly revenue at risk in the high-confidence segment

990 high-risk customers above the 75% threshold

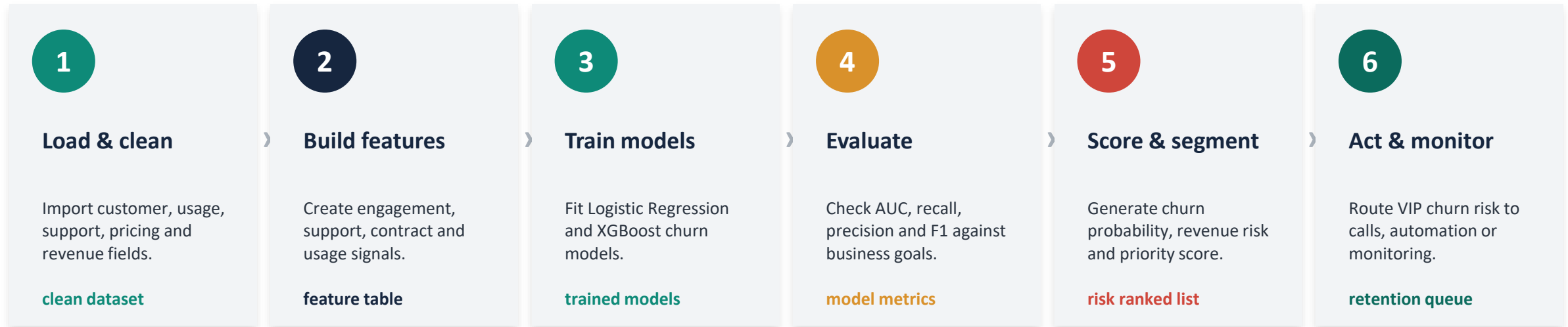
Actions now

- **Focus on high-value churners**
- **Improve product engagement**
- **Address support issues**

Decision request: launch a high-risk retention sprint this week, using the top-customer list as the daily call queue.

End-to-End Churn Modeling Process

From Python workflow to customer retention action — each code block creates a business artifact.



Feedback loop



Outreach outcomes and new behavior are logged back into the dataset, so thresholds, features and models improve over time.